Unit 13, Lesson 1: Evaluating Expressions

Benchmark

MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.

Learning Targets

- ☐ I can apply order of operations to an algebraic expression.
- ☐ I can evaluate expressions that include exponents.

13.1.1 Warm-Up: Evaluating Numeric Expressions

This math expression is dividing the internet.

What is the value of $8 \div 2(2+2)$?

1. Do you agree with Alex or Ms. B? Explain or show why using order of operations.

2. Find the value of the expression. Show your work.

$$4 - 2 \times 6 \div 3 + 7$$

13.1.2 Guided Instruction: Applying Order of Operations to Algebraic Expressions

Recall from 5th grade that an order of operations is used to evaluate numeric expressions with multiple operations. Grouping symbols such as parentheses are used to indicate which operations should be performed first. The following order is used.

Step 1: Perform operations within grouping symbols.
Step 2: Evaluate exponents.
Step 3: Multiply and divide in order from left to right $(\rightarrow)$.
Step 4: Add and subtract in order from left to right $(\rightarrow)$.

1. Three students completed the warm-up. The steps in each student's process are shown.

Leila's Work	***Fabian's Work***	***Jaylah's Work***
$4-2\times6\div3+7$ $2\times6\div3+7$ $12\div3+7$ $4+7$ 11	$4-2\times6\div3+7$ $4-12\div3+7$ $4-4+7$ $4-11$ -7	$4-2\times6\div3+7$ $4-12\div3+7$ $4-4+7$ $0+7$ 7

   a. Complete the statements about which student correctly applied the given steps of the order of operations.

○ Leila
○ Fabian
○ Jaylah

evaluated the numerical expression correctly.

She first performed all

○ multiplication and division
○ addition and subtraction

in order from left to right and then performed all

○ multiplication and division
○ addition and subtraction

in order from left to right.

   b. With your partner, discuss the work of each student, Leila, Fabian, and Jaylah. Two of these students made a mistake somewhere. Circle the mistakes that you and your partner found.

2. Omari evaluated Expression A and Expression B. His work is shown.

Expression A	***Expression B***
$12+24\div6-2\times2$ $12+4-2\times2$ $12+4-4$ $16-4$ 12	$12+24\div(6-2\times2)$ $12+24\div(6-4)$ $12+24\div(2)$ $12+12$ 24

   a. With your partner, discuss the two original expressions that Omari evaluated and his corresponding answers.

   b. Omari's partner gave him a 👍 on both of his answers. Why did Omari arrive at two different values when evaluating these expressions?

Like numeric expressions, algebraic expressions can be evaluated using order of operations when given a specific value for the ***variable***(s).

In an algebraic expression, a ***variable*** is a letter that represents a number. It is called a **vari**able because the value used to replace the letter can ***vary***.

3. When evaluating algebraic expressions, first substitute the given _________ for the variable(s). Then, apply the given steps for order of operations.

In mathematics, **evaluate** is an instruction that indicates the answer is a **value**.

4. Evaluate $\frac{n}{6}+2$ when $n=12$.

In numeric expressions, multiplication can be represented in different ways. Each expression below is a way to represent "three times five."

3×5 $\qquad$ $3\cdot 5$ $\qquad$ $3(5)$

In algebraic expressions, the multiplication symbol, $\times$, could be confused with the variable x. Therefore, writing a number next to a variable is a better way to note multiplication when using variables.

$3x$ means "three times x."

Multiplication of a number and a variable could also be represented using the multiplication dot, $3\cdot x$, or parentheses, $3(x)$, but $3x$ will be the most common way.

5. Explain why $3\cdot 5$ cannot be written as 35.

6. What is the value of $3x$ if $x=5$?

When substituting values into algebraic expressions, it is helpful to put the substituted value in **parentheses** when the variable is being multiplied by a known value.

7. Find the value of $7m-(3+b-12)\div 2$ if $m=4$ and $b=15$. Show your work.

13.1.3 Collaboration: Order of Operations

An order of operations was first noted in First Year Algebra by Webster Wells and Walter Wilson Hart. The textbook was published in 1912.

1. The problem below is an example from *First Year Algebra.*

Example. Find the numerical value of the expression,

$$4ab + \frac{6c}{b} - d^3,$$

when $a = 4, \ b = 3, \ c = 5, \ d = 2.$

   a. What is the numerical value of the expression? Show your work.

   b. Check your work with your partner.

   c. The work shown in the textbook is below.

$$4ab + \frac{6c}{b} - d^3 = 4 \cdot 4 \cdot 3 + \frac{6 \cdot 5}{3} - 2^3 = 4 \cdot 4 \cdot 3 + \frac{6 \cdot 5}{3} - 2 \cdot 2 \cdot 2$$
$$= 48 + 10 - 8 = 50.$$

   What is the difference between your work and the work from the book?

An order of operations became more important when programmers needed a set of rules to program a computer to find values of numeric expressions. In 1883, Ada Lovelace described in her book about Charles Babbage's analytical engine, the first computer, that codes could be created so the device could handle letters and symbols along with numbers.

13.1.4 Guided Instruction: Evaluating Expressions With Exponents

The work in the book, *First Year Algebra,* showed 2^3 written out as multiplication. This method can get complex as shown in the table.

$(-3)^3 + 2^6 \div -16 + 5$
Exponents written as multiplication
$((-3)(-3)(-3)) + (2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2) \div -16 + 5$

$(-3)^3 + 2^6 \div -16 + 5$
Evaluate expressions with exponents first
$-27 + 64 \div -16 + 5$

- When writing $(-3)^3$ and 2^6 using multiplication, both were written inside a set of parentheses.
- In order of operations, parentheses indicate that the operations inside should be done ___________ other steps.
- Therefore, evaluating exponents is done before other multiplication and division when following the order of operations.

1. Evaluate $(-3)^3 + 2^6 \div -16 + 5$.

2. Evaluate each expression, then determine if the expressions are equivalent.

Expression A	Expression B	Equivalent?
$(-3)^4$	$-(3^4)$	
$-(9)^2$	$(-9)^2$	
(-5^2)	$(-5)^2$	
$26 - 4^3$	$26 - (-4)^3$	
$17 + (-2^4)$	$17 - (2^4)$	

3. Describe a strategy that can be used when evaluating expressions with exponents that include a negative sign or a subtraction sign.

4. Elizabeth evaluated $b + a^3 \cdot 7 - 9$ when $a = -4$ and $b = 17$. Her work is below.

a. Circle Elizabeth's error.

$$17 + (-4)^3 \cdot 7 - 9$$
$$17 + (-64) \cdot 7 - 9$$
$$(-47) \cdot 7 - 9$$
$$-329 - 9$$
$$-338$$

b. Which step of the order of operations did she not apply correctly?

Ⓐ Evaluate expressions with exponents.
Ⓑ Perform operations inside grouping symbols.
Ⓒ Perform multiplication and division in order from left to right.
Ⓓ Perform addition and subtraction in order from left to right.

The order of performing computations is to first work within grouping symbols using the ***order of operations***. Then, simplify terms with exponents. Next, while reading from left to right, perform multiplication and division in the order in which it appears. Finally, while reading from left to right, perform addition and subtraction in the order in which it appears.

13.1.5 Try It: Discovering the Importance of Left to Right

1. Mia is evaluating the expression, $x^3 \div r \cdot (k^2 - 5)$ for $k = 3, r = -8$ and $x = 4$. Her first few steps are shown.

$$4^3 \div -8 \cdot (3^2 - 5)$$
$$64 \div -8 \cdot (9 - 5)$$
$$64 \div -8 \cdot 4$$

a. Mia thinks that she could finish her work by either finding $64 \div -8$ or $-8 \cdot 4$ first. Show the work for both possibilities to determine if Mia is correct.

b. Was Mia correct when she stated that both methods would work? Use the steps of the order of operations to justify whether or not Mia is correct.

2. Answer the following problems.

   a. Evaluate $r \cdot W \div x \div m$ when $m = 2,\ r = -4,\ W = -60,$ and $x = -10$. Show your work.

   b. Evaluate $(r \cdot W) \div (x \div m)$ when $m = 2,\ r = -4,$ $W = -60,$ and $x = -10$. Show your work.

   c. How did the use of parentheses change the answers?

13.1.6 Wrap-Up: Error Analysis

1. Nadine and Nora evaluated the expression $r^3 \div x \cdot c$ for $c = 9, r = -6$ and $x = -3$. Both students made errors in their work shown. Circle their errors.

Nadine's Work	Nora's Work
$r^3 \div x \cdot c$ $(-6)^3 \div (-3) \cdot 9$ $-216 \div (-3) \cdot 9$ $-216 \div -27$ 8	$r^3 \div x \cdot c$ $(-6)^3 \div (-3) \cdot 9$ $-18 \div (-3) \cdot 9$ $6 \cdot 9$ 54

2. Write a short explanation to each student to explain the mistake and what she needs to remember about evaluating expressions in the future.

To Nadine:	*To Nora:*